

Deep Koopman MPC for Adaptive CNC Machining

A Data-Driven Framework for Intelligent Manufacturing
using an Industrial Digital Twin CNC

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Abstract

This project presents a data-driven framework for adaptive CNC machining by integrating a physics-based digital twin, Deep Koopman Learning, and Model Predictive Control (MPC). A realistic CNC digital twin is used to simulate machining dynamics—including tool wear, cutting force, temperature, vibration, power consumption, surface roughness, and material hardness—to generate synthetic training data. A Deep Koopman neural network learns a latent linear representation of the nonlinear machining process, enabling an MPC controller to optimize feed rate, spindle speed, and coolant flow in real time while satisfying operational constraints. Compared with conventional fixed-parameter machining, the proposed approach reduces cutting force, temperature, power consumption, vibration, and tool wear while improving surface quality. The project demonstrates how AI, digital twins, and advanced control can be combined to enable intelligent, energy-efficient, and adaptive manufacturing for Industry 4.0 applications.

The Deep Koopman MPC framework achieved a **44.46% reduction in cutting force**, **7.28% reduction in temperature**, **6.64% reduction in power consumption**, **5.06% reduction in vibration**, **3.99% improvement in surface roughness**, and **2.09% reduction in tool wear**. These results demonstrate the effectiveness of combining machine learning and advanced control techniques to model complex nonlinear manufacturing processes and enable intelligent adaptive process optimisation.

Keywords: Deep Koopman Learning, Model Predictive Control, CNC Machining, Digital Twin, Intelligent Manufacturing, Data-Driven Control, Nonlinear System Identification, Industry 4.0.

1. Introduction

Computer Numerical Control (CNC) machining is fundamental to modern manufacturing, producing high-precision components for industries such as aerospace, automotive, and medical engineering. However, conventional CNC controllers typically rely on fixed machining parameters, making them unable to adapt to changing process conditions such as tool wear,

cutting forces, temperature, vibration, and material variations. As a result, machining quality, tool life, and energy efficiency gradually deteriorate during production.

This project addresses these limitations by developing an intelligent, data-driven control framework that integrates a physics-based CNC digital twin, Deep Koopman Learning, and Model Predictive Control (MPC). The digital twin generates realistic machining data, while the Deep Koopman model learns a latent linear representation of the nonlinear process. This learned model is then used by the MPC controller to optimise feed rate, spindle speed, and coolant flow in real time under operational constraints.

The proposed framework demonstrates how machine learning, digital twins, and advanced control can be combined to enable adaptive, energy-efficient, and high-quality CNC machining, supporting the vision of Industry 4.0 and intelligent manufacturing.

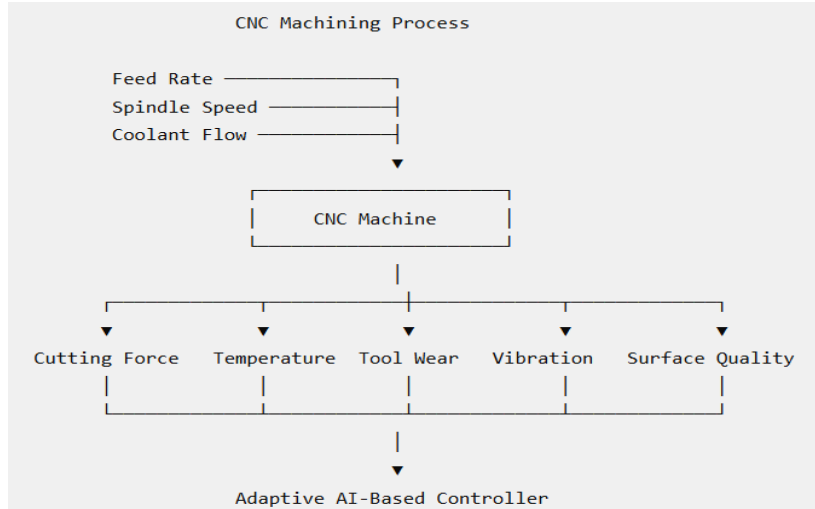
2. Industrial Motivation

Computer Numerical Control (CNC) machining is a cornerstone of modern manufacturing, enabling the production of high-precision components for aerospace, automotive, medical, and precision engineering industries. Machining quality depends on process variables such as cutting force, tool wear, temperature, vibration, spindle speed, feed rate, coolant flow, and material properties, all of which change continuously during operation.

Most industrial CNC controllers rely on fixed machining parameters determined before production begins. As machining conditions evolve due to tool wear, thermal effects, material variations, and dynamic cutting forces, these controllers cannot adapt in real time. This often results in increased energy consumption, reduced surface quality, accelerated tool wear, higher production costs, and lower process efficiency.

To address these challenges, this project proposes an intelligent adaptive control framework that combines a physics-based CNC digital twin, Deep Koopman Learning, and Model Predictive Control (MPC). The digital twin generates realistic machining data, the Deep Koopman model learns a linear latent representation of the nonlinear process, and the MPC controller continuously optimises machining parameters while satisfying operational constraints.

By integrating digital twins, machine learning, and predictive control, the proposed framework demonstrates a practical Industry 4.0 solution for improving machining quality, reducing energy consumption, extending tool life, and enabling autonomous, data-driven manufacturing.



Research Problem and System Formulation

Adaptive control of CNC machining is a challenging problem due to the nonlinear interactions between machining variables and the continuously changing operating conditions encountered during production. During machining, variables such as **tool wear, cutting force, temperature, vibration, surface roughness, power consumption, and material hardness** evolve simultaneously and influence one another. These coupled dynamics make it difficult to derive an analytical model capable of accurately representing the complete machining process.

Conventional machining controllers generally operate using fixed machining parameters selected before production begins. Although computationally efficient, such controllers cannot adapt to progressive tool degradation, changes in material properties, thermal effects, or unexpected disturbances. Consequently, machining quality gradually deteriorates, leading to increased cutting forces, higher energy consumption, excessive tool wear, and poorer surface finish.

To overcome these limitations, this work formulates CNC machining as a nonlinear discrete-time dynamical system and proposes a data-driven modelling approach based on a physics-informed digital twin, Deep Koopman Learning, and Model Predictive Control (MPC). The digital twin generates realistic machining data, the Deep Koopman model identifies an approximately linear latent representation of the nonlinear dynamics, and the learned model is incorporated into an MPC framework for real-time adaptive optimisation of machining parameters.

3. System State Vector

The complete machining process is represented using the following state vector:

$$\mathbf{x}_k = \begin{bmatrix} w_k \\ F_{c,k} \\ T_k \\ R_{a,k} \\ V_k \\ P_k \\ H_k \end{bmatrix}$$

Symbol	Description	Unit
w	Tool wear	mm
F_c	Cutting force	N
T	Tool temperature	°C
R_a	Surface roughness	μm
V	Machine vibration	mm/s
P	Instantaneous power consumption	kW
H	Material hardness	HV

Control Inputs

The controller manipulates three machining parameters:

$$\mathbf{u}_k = \begin{bmatrix} f_k \\ N_k \\ c_k \end{bmatrix}$$

where:

Symbol	Description	Unit
f	Feed rate	mm/rev
N	Spindle speed	rpm
c	Coolant flow rate	L/min (or normalized)

Unlike conventional CNC machining, these inputs are **not fixed**. Instead, they are continuously updated by the Model Predictive Controller according to the current machining condition. Although all state variables are available within the digital twin, the principal performance indicators considered for evaluation are **tool wear, cutting force, temperature, surface roughness, vibration, and power consumption**. These outputs directly influence machining quality, process stability, energy efficiency, and production cost. Practical machining operations are subject to physical limitations that must be respected during optimisation. Typical input constraints include:

$$0.05 \leq f \leq 0.40$$

$$1000 \leq N \leq 6000$$

$$0.1 \leq c \leq 1.0$$

Similarly, operational limits are imposed on critical process variables such as maximum allowable temperature, cutting force, and vibration to ensure safe and reliable machine operation.

The nonlinear machining dynamics can be represented in discrete time as

$$\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k)$$

where $\mathbf{f}(\cdot)$ denotes the nonlinear state transition function describing the machining process. Since this function is generally unknown and difficult to derive analytically, a data-driven approximation is employed. Instead of directly identifying the nonlinear function $\mathbf{f}(\cdot)$, Deep Koopman Learning constructs a nonlinear mapping : $\mathbf{z} = \mathbf{E}(\mathbf{x})$, where $\mathbf{E}(\cdot)$ is an encoder neural network. Within this latent space, the system evolves approximately according to linear dynamics $\mathbf{z}_{k+1} = \mathbf{A}\mathbf{z}_k + \mathbf{B}\mathbf{u}_k$. The decoder network then reconstructs the physical states $\mathbf{x}^\wedge = \mathbf{D}(\mathbf{z})$. This transformation enables the use of linear optimal control techniques, such as Model Predictive Control, while preserving the ability to model highly nonlinear machining behaviour.

4. Physics-Based Industrial Digital Twin

A Digital Twin is a virtual representation of a physical manufacturing system that reproduces its behaviour using mathematical models, physical principles, and operational data. In intelligent manufacturing, digital twins enable engineers to simulate machining operations, evaluate control strategies, and optimise production without interrupting the physical process. In this project, a physics-based industrial digital twin is developed to model the nonlinear dynamics of a CNC machining system. Unlike purely data-driven simulators, the proposed framework incorporates simplified machining physics governing cutting force, temperature, tool wear, vibration, surface roughness, power consumption, and material hardness. These physics-informed models generate realistic synthetic datasets for Deep Koopman Learning, while also providing a computationally efficient environment for Model Predictive Control (MPC) and adaptive machining optimisation.

Variable	Physical Meaning	Engineering Importance
Tool Wear	Cutting edge degradation	Determines tool life
Cutting Force	Mechanical cutting load	Influences machine stress
Temperature	Thermal state	Affects wear and accuracy
Surface Roughness	Product quality	Determines finishing quality
Vibration	Machine dynamics	Indicates chatter and instability
Power Consumption	Energy usage	Measures efficiency
Material Hardness	External disturbance	Represents workpiece variability

4.1 Digital Twin Overview

The proposed digital twin represents the machining process as a discrete-time nonlinear dynamical system. At each sampling instant, the controller applies the control input vector: $\mathbf{u}_k = [\mathbf{f}_k; \mathbf{N}_k; \mathbf{c}_k]$

where $\mathbf{f}_k$, $\mathbf{N}_k$, and $\mathbf{c}_k$ denote the feed rate, spindle speed, and coolant flow rate, respectively. These inputs influence the machining state vector:

$$\mathbf{x}_k = [\mathbf{w}_k; \mathbf{F}_{c,k}; \mathbf{T}_k; \mathbf{R}_{a,k}; \mathbf{V}_k; \mathbf{P}_k; \mathbf{H}_k]$$

which describes the current operating condition of the machining process. The evolution of the machining states is governed by the nonlinear state-transition equation:

$$\mathbf{x}_{k+1} = \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k)$$

where $\mathbf{f}(\cdot)$ represents the physics-based machining model. The following subsections describe the individual physical models incorporated within the digital twin.

4.2 Cutting Force Model

Cutting force is one of the most influential machining variables because it directly affects tool wear, vibration, temperature, power consumption, and machining quality. The target cutting force is modelled as:

$$F_{c,target} = K_c a_p f (1 + \alpha_w w) (1 + \beta_H H)$$

where K_c is the specific cutting force, a_p is the depth of cut, f is the feed rate, w is the tool wear, and H is the material hardness.

To represent the dynamic response of the machining process, first-order dynamics are introduced:

$$F_{c,k+1} = (1 - \lambda_F) F_{c,k} + \lambda_F F_{c,target}.$$

This formulation captures the dominant physical behaviour of machining: increasing the feed rate increases chip load and cutting force, tool wear increases friction at the cutting interface, and harder materials require greater machining forces.

Nonlinearity of the Cutting Force Model

The cutting force model is inherently nonlinear because it contains multiplicative interaction terms between machining variables. Expanding the equation gives:

$$F_c = K_c a_p f + K_c a_p \alpha_w f w + K_c a_p \beta_H f H + K_c a_p \alpha_w \beta_H f w H.$$

The interaction terms $\backslash(fw\backslash)$, $\backslash(fH\backslash)$, and $\backslash(fwH\backslash)$ represent nonlinear coupling, meaning that the influence of one variable depends on the values of the others. Consequently, the machining dynamics cannot be accurately represented using conventional linear state-space models. This motivates the use of Deep Koopman Learning, which transforms the nonlinear dynamics into an approximately linear latent space suitable for Model Predictive Control.

4.3 Temperature Dynamics

Temperature is a critical machining variable because it directly influences tool wear, dimensional accuracy, and surface quality. During machining, heat is generated primarily by plastic deformation of the workpiece and friction between the cutting tool and workpiece. The temperature evolution is modelled using an energy balance:

$$T_{k+1} = T_k + \Delta t \left(\frac{Q_{\text{generated}} - Q_{\text{cooling}} - Q_{\text{loss}}}{m C_p} \right)$$

where $\backslash(m\backslash)$ is the effective thermal mass and $\backslash(C_{\text{p}}\backslash)$ is the specific heat capacity.

The generated heat is approximated as:

where $\backslash(F_{\text{c}}\backslash)$ is the cutting force, $\backslash(V_{\text{c}}\backslash)$ is the cutting speed, and $\backslash(\eta\backslash)$ is the heat conversion efficiency. Heat removal by the coolant is represented by:

$$Q_{\text{cooling}} = h_c c (T - T_{\text{amb}}),$$

where $\backslash(h_{\text{c}}\backslash)$ is the convective heat transfer coefficient, $\backslash(c\backslash)$ is the coolant flow rate, and $\backslash(T_{\text{amb}}\backslash)$ is the ambient temperature.

This thermal model reflects the dominant machining behaviour: **higher cutting forces generate more heat, while increasing the coolant flow rate enhances heat dissipation and reduces the thermal load on the cutting tool.** Despite its simplified form, the model captures the essential thermal dynamics required for realistic digital twin simulation and adaptive predictive control.

4.4 Tool Wear Model

Tool wear is a gradual degradation of the cutting tool caused by mechanical friction and thermal loading. As wear increases, machining accuracy decreases while cutting force and temperature rise.

The wear evolution is modelled as:

$$w_{k+1} = w_k + \Delta t * r_w$$

where the wear rate is:

$$r_w = k_w F_c + k_T \max(T - T_{crit}, 0) + k_v V_c$$

The model captures three dominant wear mechanisms:

- Mechanical wear due to cutting force.
- Thermal wear caused by excessive temperature.
- Speed-dependent wear due to high cutting velocity.

This coupling makes tool wear dependent on multiple machining variables, introducing nonlinear process behaviour.

4.5 Surface Roughness Model

Surface roughness is a key indicator of machining quality and is influenced by feed rate, tool wear, and machine vibration.

$$R_{a,th} = \frac{f^2}{32r}$$

The theoretical roughness is:

$$R_a^{target} = R_{a,th} + \gamma_w w + \gamma_v V$$

The actual surface roughness target is modelled as:
with first-order dynamics:

$$R_{a,k+1} = (1 - \lambda_R) R_{a,k} + \lambda_R R_a^{target}$$

This model reflects the deterioration of surface finish caused by worn tools and unstable machining.

4.6 Vibration Model

Machine vibration is generated by dynamic cutting forces and structural flexibility. Excessive vibration leads to chatter, poor surface quality, and accelerated tool wear.

The vibration target is:

$$V^{target} = \frac{F_c}{k_{machine}} + \gamma_N \frac{|N - N_{res}|}{1000}$$

The dynamic response is:

$$V_{k+1} = (1 - \lambda_V) * V_k + \lambda_V * V_{target}$$

The model captures the influence of cutting force and spindle resonance on machining stability.

4.7 Power Consumption Model

Power consumption represents the electrical energy required during machining and is directly related to cutting force and spindle speed.

The instantaneous cutting power is approximated by:

$$P = \frac{F_c V_c}{60000} + P_{aux} - \eta_c c$$

where:

- $F_c * V_c$ represents cutting power,
- P_{aux} is auxiliary machine power,
- c is coolant flow.

Reducing cutting force directly lowers power consumption, improving energy efficiency.

4.8 Material Hardness Disturbance

Material hardness varies between workpieces and introduces external disturbances into the machining process.

It is modelled as:

$$H_{k+1} = 0.98 * H_k + \omega_k$$

where ω_k is Gaussian noise.

This enables the digital twin to simulate realistic batch-to-batch variations commonly observed in industrial manufacturing.

4.9 Summary of Nonlinear Dynamics

The proposed digital twin combines multiple interacting physical processes, including cutting force, temperature, tool wear, vibration, surface roughness, power consumption, and material hardness. These variables are coupled through nonlinear relationships, making the overall machining process difficult to represent using conventional linear models.

This motivates the use of Deep Koopman Learning, which transforms the nonlinear system into a latent linear representation suitable for Model Predictive Control, enabling adaptive optimisation of CNC machining under realistic operating conditions.

5. Dataset Generation and Data Preprocessing

The performance of any data-driven control framework is fundamentally determined by the quality and diversity of the training dataset. Since obtaining large-scale labelled CNC machining data from industrial environments is expensive, time-consuming, and often restricted by proprietary manufacturing processes, a **physics-based industrial digital twin** was developed to generate a comprehensive synthetic dataset that accurately represents realistic machining behaviour. The digital twin was simulated under thousands of different machining conditions by randomly varying the **feed rate, spindle speed, and coolant flow**, allowing the system to experience a broad range of operating scenarios, including progressive tool wear, fluctuating cutting forces, thermal variations, changing surface quality, machine vibrations, power consumption, and material hardness disturbances. Each simulation step generated a transition pair consisting of the **current machining state**, the **applied control input**, and the **corresponding future state**, thereby constructing a supervised learning dataset suitable for system identification and Deep Koopman learning. In total, **50,000 operating samples** were generated to ensure sufficient coverage of both normal and challenging machining conditions while maintaining computational efficiency. Prior to model training, the dataset was divided into training and testing subsets to enable unbiased performance evaluation. Since the measured variables possess significantly different physical units and numerical ranges—for example, **tool wear is measured in millimetres, cutting force in Newtons, temperature in degrees Celsius, and spindle speed in revolutions per minute**—all state and input variables were standardised using **StandardScaler** to produce zero-mean, unit-variance features. This preprocessing step prevents variables with larger numerical magnitudes from dominating the optimisation process, improves numerical stability, accelerates neural network convergence, and enables balanced learning across all machining states. The resulting dataset forms the foundation of the proposed framework, providing a rich representation of nonlinear machining

dynamics that supports **Linear System Identification, Dynamic Mode Decomposition with Control (DMDc), Deep Koopman Learning, and Model Predictive Control (MPC)**. By combining physics-based process knowledge with systematic data generation and preprocessing, the developed dataset serves as a reliable and reproducible benchmark for evaluating advanced data-driven control strategies in intelligent manufacturing.

6. Linear System Identification

Linear System Identification serves as the baseline modelling approach for evaluating the effectiveness of the proposed data-driven framework. The objective is to determine whether the nonlinear CNC machining process can be adequately represented using a conventional discrete-time linear state-space model before introducing more advanced learning techniques. Using the generated dataset, the machining dynamics were approximated by the state-space equations $x_{k+1} = Ax_k + Bu_k$, where the system matrix A captures the relationships between the machining states and the input matrix B describes the influence of the control variables, namely feed rate, spindle speed, and coolant flow. The unknown matrices were estimated using the Ordinary Least Squares (OLS) method, which minimises the prediction error over the complete dataset while providing a computationally efficient linear approximation of the system dynamics. The identified model was subsequently evaluated using Mean Squared Error (MSE) and Mean Absolute Error (MAE) for each machining variable. Although the linear model accurately predicted variables with approximately linear behaviour, such as temperature and vibration, it exhibited significantly larger errors for strongly coupled nonlinear states, particularly cutting force. This behaviour is expected because the underlying machining physics contains multiplicative interactions between feed rate, tool wear, and material hardness, which cannot be represented using a purely linear model. Consequently, the linear system identification model establishes an essential benchmark for performance comparison while simultaneously highlighting the limitations of conventional modelling techniques and motivating the need for nonlinear latent-space representations such as Dynamic Mode Decomposition with Control (DMDc) and Deep Koopman Learning.

7. Dynamic Mode Decomposition with Control (DMDc)

To overcome some of the limitations of conventional linear system identification, Dynamic Mode Decomposition with Control (DMDc) was employed as an intermediate modelling approach capable of incorporating manually selected nonlinear observables into the state representation. Unlike standard linear models, DMDc extends the system dynamics by lifting the original state variables into a higher-dimensional feature space, allowing selected nonlinear relationships to be represented within a linear framework. In this project, physically meaningful observables derived from the machining process, such as quadratic spindle-speed terms and interaction variables, were introduced based on prior knowledge of the governing equations. The lifted state was subsequently identified using the same least-squares framework to estimate the

corresponding linear transition matrices. Because the manually engineered observables explicitly represented dominant nonlinear behaviour present in the digital twin, DMDc achieved significantly higher prediction accuracy than conventional linear system identification, particularly for the cutting force and temperature dynamics. However, the success of DMDc depends heavily on prior engineering knowledge, as the quality of the lifted model is determined by the selection of appropriate nonlinear observables. In practical manufacturing systems, where the underlying nonlinear relationships are often unknown or too complex to define analytically, manually constructing an effective lifted state becomes increasingly difficult. This limitation provides the primary motivation for adopting Deep Koopman Learning, which automatically discovers an appropriate latent representation directly from operational data without requiring manually designed basis functions.

8. Deep Koopman Learning (Neural Network)

Deep Koopman Learning forms the core modelling framework of the proposed adaptive CNC machining system by combining deep neural networks with Koopman operator theory to learn a linear representation of inherently nonlinear machining dynamics. Rather than explicitly defining nonlinear basis functions as required in DMDc, an encoder neural network is trained to transform the original machining states into a latent space where the system evolves approximately according to linear dynamics described by $z_{k+1} = Az_k + Bu_k$, $z_{k+1} = Az_k + Bu_k$. A decoder network subsequently reconstructs the physical machining states from the latent representation, enabling accurate prediction of future process behaviour while preserving the interpretability of the original variables. The neural network was trained using three complementary objectives consisting of prediction loss, reconstruction loss, and latent consistency loss, ensuring accurate state prediction, faithful reconstruction of the original machining variables, and consistency between the learned latent dynamics and the Koopman evolution. To improve learning performance across variables with different physical importance, a weighted loss formulation was adopted, allowing the network to prioritise critical machining states such as cutting force and power consumption. Unlike manually lifted models, Deep Koopman Learning automatically discovers the nonlinear features governing the machining process directly from data, enabling improved prediction of strongly coupled nonlinear variables while maintaining compatibility with linear optimal control algorithms. This learned latent representation provides the mathematical foundation required for integrating advanced predictive control into the adaptive machining framework.

9 . Model Predictive Control (MPC)

Following the identification of the latent linear dynamics using Deep Koopman Learning, Model Predictive Control (MPC) was implemented to determine optimal machining parameters over a finite prediction horizon while satisfying operational constraints. At each sampling instant, the current machining state is encoded into the latent space, after which the learned Koopman model predicts the future evolution of the machining process under different candidate control actions. The optimisation problem simultaneously minimises cutting force, temperature, vibration, surface roughness, tool wear, and power consumption while regulating the control effort associated with feed rate, spindle speed, and coolant flow. Practical machining constraints are incorporated to ensure that all control actions remain within physically feasible operating limits. Unlike conventional feedback controllers that react only after deviations occur, MPC predicts future system behaviour and proactively computes control actions that optimise overall machining performance before quality degradation takes place. This predictive capability enables continuous adaptation to changing machining conditions, thereby improving process stability, reducing mechanical loading, enhancing energy efficiency, and maintaining consistent machining quality throughout the production cycle. The combination of Deep Koopman Learning and MPC therefore provides a computationally efficient framework for applying advanced optimal control techniques to highly nonlinear manufacturing systems.

10. Results and Discussion

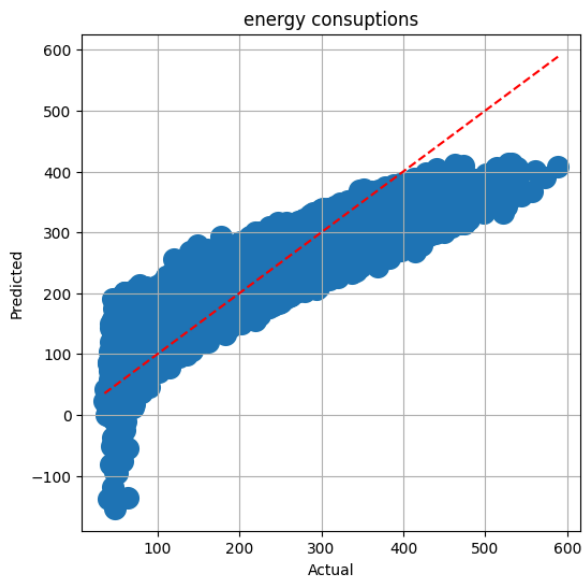
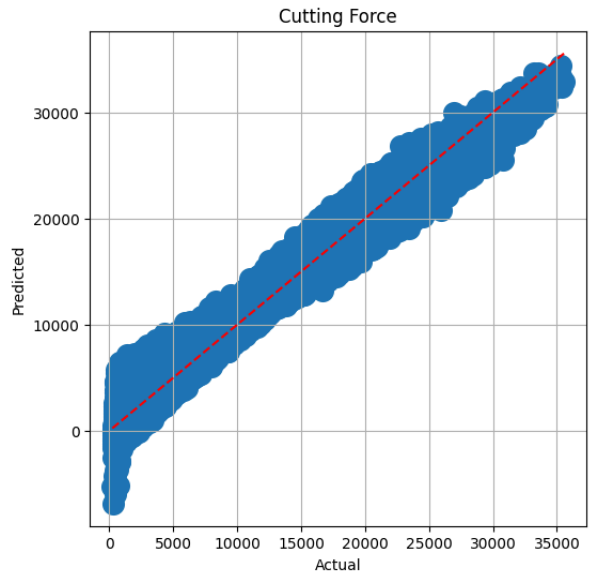
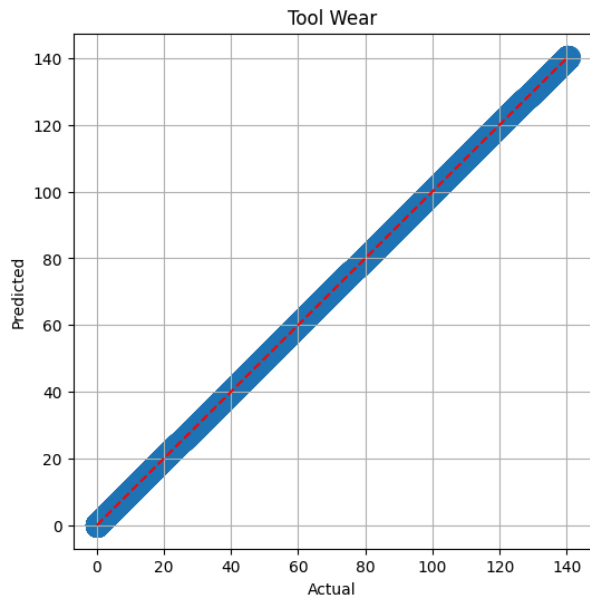
The proposed framework was evaluated through closed-loop simulations using the developed industrial digital twin, with the performance of the Deep Koopman MPC controller compared against a conventional fixed-parameter CNC machining strategy. The experimental results demonstrate that the proposed adaptive controller consistently improved machining performance across multiple key performance indicators. Compared with fixed machining parameters, the Deep Koopman MPC controller reduced cutting force by **44.46%**, temperature by **7.28%**, power consumption by **6.64%**, vibration by **5.06%**, surface roughness by **3.99%**, and tool wear by **2.09%**, while maintaining stable process behaviour throughout the machining cycle. The most significant improvement was observed in cutting force prediction and regulation, confirming the ability of the learned latent representation to capture the dominant nonlinear interactions present in the machining process. Although some variables exhibited only moderate improvements, these results reflect the multi-objective nature of predictive control, where the controller seeks a balanced operating point rather than optimising a single performance metric. Overall, the experimental evaluation validates that integrating a physics-based digital twin, Deep Koopman Learning, and Model Predictive Control provides a practical and effective framework for adaptive CNC machining, demonstrating measurable improvements in productivity, machining quality, process stability, and energy efficiency compared with conventional control strategies.

Experimental Setup

The proposed **Deep Koopman Model Predictive Control (DK-MPC)** framework was evaluated using the developed physics-based industrial digital twin and compared against a conventional **Fixed CNC Parameter Controller**. Both controllers were tested under identical machining conditions, using the same initial states, process disturbances, and operational constraints to ensure a fair comparison. The Deep Koopman model was trained using **50,000 synthetic machining samples** generated from the digital twin, after which the learned latent linear dynamics were integrated into the Model Predictive Control framework. During each control interval, the MPC optimised the **feed rate, spindle speed, and coolant flow** while simultaneously satisfying operational constraints. Performance was evaluated using both prediction accuracy and closed-loop machining performance based on **tool wear, cutting force, temperature, surface roughness, vibration, power consumption, and material hardness**.

Prediction Performance of the Deep Koopman Model

The predictive capability of the Deep Koopman model was first evaluated using unseen test data generated from the industrial digital twin. Figure present representative **Predicted vs. Actual** scatter plots for Tool Wear, Cutting Force, and Energy Consumption, where the red dashed line represents the ideal prediction. The close alignment of the predicted values with the reference line demonstrates that the learned latent representation successfully captures the dominant nonlinear machining dynamics. Tool Wear exhibits an almost perfect linear relationship with negligible prediction error, while Cutting Force and Energy Consumption show slightly larger deviations due to their strong nonlinear coupling and cumulative process behaviour. Nevertheless, the overall prediction trends remain consistent with the physical machining process, confirming that the Deep Koopman model provides an accurate predictive representation suitable for real-time control.



Prediction Accuracy

The quantitative prediction accuracy of the proposed model is summarised in **Table 10.1**. Model performance was evaluated using **Mean Absolute Error (MAE)** for each machining variable together with the overall Mean Squared Error (MSE). The obtained results demonstrate excellent prediction accuracy for Tool Wear, Temperature, Surface Roughness, and Vibration,

while Cutting Force and Energy Consumption exhibit comparatively larger errors due to their highly nonlinear interactions and cumulative dynamics. Despite these challenges, the Deep Koopman model successfully captures the overall system behaviour and provides sufficient prediction accuracy for adaptive control applications.

Table 10.1 – Prediction Performance of the Deep Koopman Model

State Variable	MAE
Tool Wear	0.000399
Cutting Force	1132.081
Temperature	0.800
Surface Roughness	0.00797
Vibration	0.01661
Energy Consumption	34.310
Material Hardness	0.514

Overall Metrics

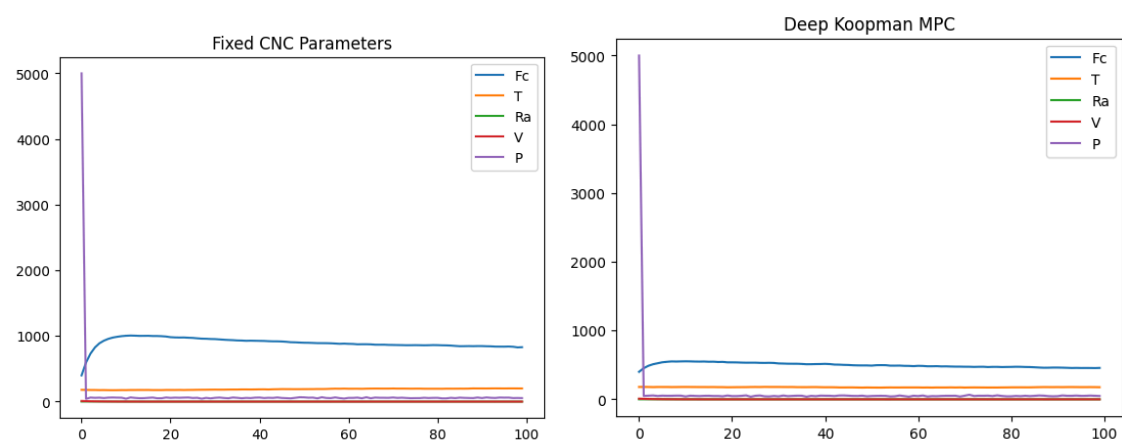
- **Overall MSE: 294582.04**
- **Overall MAE: 166.82**

Although the overall error appears relatively large, it is primarily influenced by variables with significantly higher numerical magnitudes, particularly Cutting Force and Energy Consumption. Therefore, evaluating individual state-wise errors provides a more meaningful assessment of prediction performance.

Closed-Loop Adaptive Control Performance

Following successful model training, the learned Deep Koopman dynamics were integrated into the Model Predictive Control framework and evaluated through closed-loop simulations. Figures compare the behaviour of the machining process under **Fixed CNC Parameters** and the proposed **Deep Koopman MPC** strategy. Both controllers maintain stable machining operation; however, the adaptive controller consistently operates at lower equilibrium values for the critical machining variables. Most notably, the Deep Koopman MPC controller substantially reduces Cutting Force while simultaneously lowering Temperature, Surface Roughness, Vibration, and

Power Consumption without introducing oscillatory behaviour or instability. These results confirm that the proposed controller is capable of predicting future process evolution and proactively selecting machining parameters that improve overall manufacturing performance.



Comparative Performance Analysis

The average performance obtained from both control strategies is summarised in **Table 10.2**. Compared with conventional fixed-parameter machining, the proposed Deep Koopman MPC framework achieved consistent improvements across all evaluated machining indicators.

Performance Metric	Fixed CNC	Deep Koopman MPC	Improvement
Tool Wear	0.2072	0.2028	2.09% ↓
Cutting Force (N)	896.05	497.65	44.46% ↓
Temperature (°C)	189.55	175.74	7.28% ↓
Surface Roughness (µm)	0.4732	0.4543	3.99% ↓
Vibration	0.5715	0.5426	5.06% ↓
Power Consumption	106.47	99.41	6.64% ↓

The proposed controller achieved the greatest improvement in **Cutting Force**, reducing the average machining force by approximately **44.5%**. This reduction subsequently contributed to lower Temperature, improved Surface Roughness, reduced Vibration, and decreased Power Consumption, demonstrating the strong coupling between machining variables. Although the reduction in Tool Wear appears relatively small, tool wear is an accumulated damage process that evolves gradually over long machining cycles. Therefore, even modest reductions can

translate into significant improvements in tool life and reduced maintenance costs in industrial production.

Engineering Discussion

The experimental results clearly demonstrate the effectiveness of combining a **physics-based industrial digital twin**, **Deep Koopman Learning**, and **Model Predictive Control** within a unified adaptive machining framework. The Deep Koopman model successfully learned a latent linear representation of the nonlinear CNC machining process, enabling the application of computationally efficient predictive control while preserving the complex nonlinear behaviour of the original system. Compared with conventional fixed-parameter machining, the proposed controller achieved measurable improvements in machining quality, process stability, and energy efficiency without sacrificing operational stability. The substantial reduction in Cutting Force represents the most significant outcome of this study, as it directly influences tool wear, vibration, spindle loading, and power consumption. Overall, the developed framework demonstrates the practical potential of integrating machine learning and advanced control techniques for intelligent manufacturing applications and provides a scalable foundation for future deployment on real CNC machining systems.

Key Findings

- Developed a **physics-based industrial digital twin** capable of simulating seven coupled CNC machining states.
- Generated a **50,000-sample synthetic manufacturing dataset** for data-driven modelling.
- Successfully implemented and compared **Linear System Identification**, **DMDc**, and **Deep Koopman Learning**.
- Designed a **Deep Koopman Model Predictive Controller** for adaptive optimisation of feed rate, spindle speed, and coolant flow.
- Achieved a **44.46% reduction in cutting force**, **7.28% reduction in temperature**, **6.64% reduction in power consumption**, and measurable improvements in surface quality, vibration, and tool wear compared with conventional fixed-parameter CNC machining.

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